OS ORANGE PROBLEM

Phase1

```
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
test_objects: test_objects.c:39: test_blob_storage: Assertion `rc == 0' failed.
Aborted (core dumped)
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

```
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_read':
object.c:218:5: warning: ignoring return value of 'fread' declared with attribute 'warn_u
  218 |     fread(buffer, 1, size, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~
object.c: In function 'object_write':
object.c:147:47: warning: ".tmp" directive output may be truncated writing 4 bytes into a
12 [-Wformat-truncation=]
  147 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                               ^~~~
In file included from /usr/include/stdio.h:980,
                 from pes.h:8,
                 from object.c:11:
In function 'snprintf',
    inlined from 'object_write' at object.c:147:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:54:10: note: '__builtin___snprintf_chk' outpu
a destination of size 512
   54 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   55 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   56 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
PASS: blob storage
PASS: deduplication
test_objects: test_objects.c:80: test_integrity: Assertion `rc == -1' failed.
Aborted (core dumped)
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

Phase 2

```
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

Phase 3

```
nethra@nethra-VirtualBox: ~/PES2UG24AM101-pes-vcs
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes add file1.txt file2.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.h
  untracked:  test_objects.c
  untracked:  test_tree.c
  untracked:  .gitignore
  untracked:  tree.h
  untracked:  pes.c
  untracked:  Makefile
  untracked:  README.md
  untracked:  object.c
  untracked:  commit.c
  untracked:  pes.h
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  commit.h
  untracked:  index.c
```

```
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776408660 6 file
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776408676 6 file
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

Phase 4

```
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ echo "Hello" > hello.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes add hello.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes commit -m "Initial commit"
Committed: 7fe8f6122308... Initial commit
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ echo "World" >> hello.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes add hello.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes commit -m "Add world"
Committed: 56c4a1ec212d... Add world
Help
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ echo "Bye" > bye.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes add bye.txt
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes commit -m "Add farewell"
Committed: 4c080ffbbb8d... Add farewell
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ ./pes log
commit 4c080ffbbb8d520b436633460d2da4e8a731501b7910fd44abe501cb7f76ca02
Author: PES User <pes@localhost>
Date:   1776409962

    Add farewelldd f

commit 56c4a1ec212d490970d9c7d6d047da2f2f679c8a2d507102765c7b63f1c4bd8a
Author: PES User <pes@localhost>
Date:   1776409924

    Add worldelldd f

commit 7fe8f612230889d7a2cc44b360535f2efe5bbdefa871fc378f93ac690c5b6dd1
Author: PES User <pes@localhost>
Date:   1776409866

    Initial commit

nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

```
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/4c/080ffbbb8d520b436633460d2da4e8a731501b7910fd44abe501cb7f76ca02
.pes/objects/56/c4a1ec212d490970d9c7d6d047da2f2f679c8a2d507102765c7b63f1c4bd8a
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/69/83b0e291beaa233851ea57e9a9bd21cfd42328e358091362c592ed075dd9b9
.pes/objects/7f/e8f612230889d7a2cc44b360535f2efe5bbdefa871fc378f93ac690c5b6dd1
.pes/refs/heads/main
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

```
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ cat .pes/refs/heads/main
4c080ffbbb8d520b436633460d2da4e8a731501b7910fd44abe501cb7f76ca02
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

Final

```
4c080ffbbb8d520b436633460d2da4e8a731501b7910fd44abe501cb7f76ca02
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: a1ac2c3d91c7... Initial commit

Log after first commit:
commit a1ac2c3d91c7b7a00edccc33ecaafe2fb6d0b94eee7b5fa8416e30ebdd472de2
Author: PES User <pes@localhost>
```

```
Author: PES User <pes@localhost>
Date:   1776410205

    Update file.txtf

commit a1ac2c3d91c7b7a00edccc33ecaafe2fb6d0b94eee7b5fa8416e30ebdd472de2
Author: PES User <pes@localhost>
Date:   1776410205

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
448d3fac5701f00d078d999bb5b724cf9fed05c3525d1b7fa53fadc34eacf5a2

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/32/590cb96e2f0bed7b415ae54a9f675d73bdcf68dcefe384afc9534f39dd802d
.pes/objects/44/8d3fac5701f00d078d999bb5b724cf9fed05c3525d1b7fa53fadc34eacf5a2
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/69/83b0e291beaa233851ea57e9a9bd21cfd42328e358091362c592ed075dd9b9
.pes/objects/a1/ac2c3d91c7b7a00edccc33ecaafe2fb6d0b94eee7b5fa8416e30ebdd472de2
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
nethra@nethra-VirtualBox:~/PES2UG24AM101-pes-vcs$
```

Q5.1 Branching and Checkout

To implement pes checkout <branch>, first check if the branch file exists in .pes/refs/heads/. Then update .pes/HEAD to point to that branch. After that, read the commit hash from the branch file, load its tree object, and update the working directory files to match that snapshot.

This operation is complex because local modified or untracked files may be overwritten during checkout.

Q5.2 Dirty Working Directory Conflict Detection

Use the index and object store to compare tracked files.

- If a file in the working directory differs from the index, it has uncommitted changes.
- If the same file is different in the target branch, checkout must stop.

This prevents loss of local changes.

Q5.3 Detached HEAD

Detached HEAD means .pes/HEAD stores a commit hash directly instead of a branch reference.

Commits can still be created, but no branch points to them. If the user switches branches later, those commits may become unreachable.

Recovery is possible by creating a new branch pointing to that commit hash.

Q6.1 Garbage Collection

Start from all references (HEAD, branches) and mark reachable objects.

Traverse:

- Commit → parent + tree
- Tree → subtrees + blobs
- Blob → mark only

Use a hash set to track visited hashes.

After traversal, delete all unmarked objects from .pes/objects/.

For 100,000 commits and 50 branches, several hundred thousand objects may need to be checked.

Q6.2 Concurrent GC Danger

Running garbage collection during a commit is dangerous.

Example:

1. Blob object is written
2. GC sees no reference yet and deletes it
3. Commit later points to missing blob

This can corrupt the repository.

Git avoids this using locks, temporary protection, and safe GC timing.